Program 10

Title: Demonstrate Kubernetes Dashboard using CLI and Kubernetes Cluster
Objective: Deploy an Nginx application, expose it via a service, and monitor the cluster using the Kubernetes Dashboard.

Project Structure

```
new-program-10/
│
├── deployment.yaml
└── services.yaml
```

Configuration Files

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

services.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
```

Execution Steps

Step 1: Project Initialization

```
1rv24mc078_prateek_kumar@arch mkdir new-program-10
1rv24mc078_prateek_kumar@arch cd new-program-10
```

Step 2: Apply Deployment

```
1rv24mc078_prateek_kumar@arch ~/new-program-10 kubectl apply -f
deployment.yaml
deployment.apps/nginx-deployment created
```

Step 3: Apply Service

```
1rv24mc078_prateek_kumar@arch ~/new-program-10 kubectl apply -f
services.yaml
service/nginx-service created
```

Step 4: Verify Pods

```
1rv24mc078_prateek_kumar@arch kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
nginx-deployment-6f96d4446b-a12cd   1/1     Running   0          2m
```

```
nginx-deployment-6f96d4446b-b34ef   1/1     Running   0          2m
```

Step 5: Verify Services

```
1rv24mc078_prateek_kumar@arch kubectl get services
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)
AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP
10m
nginx-service   NodePort    10.105.91.3     <none>        80:32649/TCP   2m
```

Step 6: Verify Deployment

```
1rv24mc078_prateek_kumar@arch kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           2m
```

Step 7: Access the Application

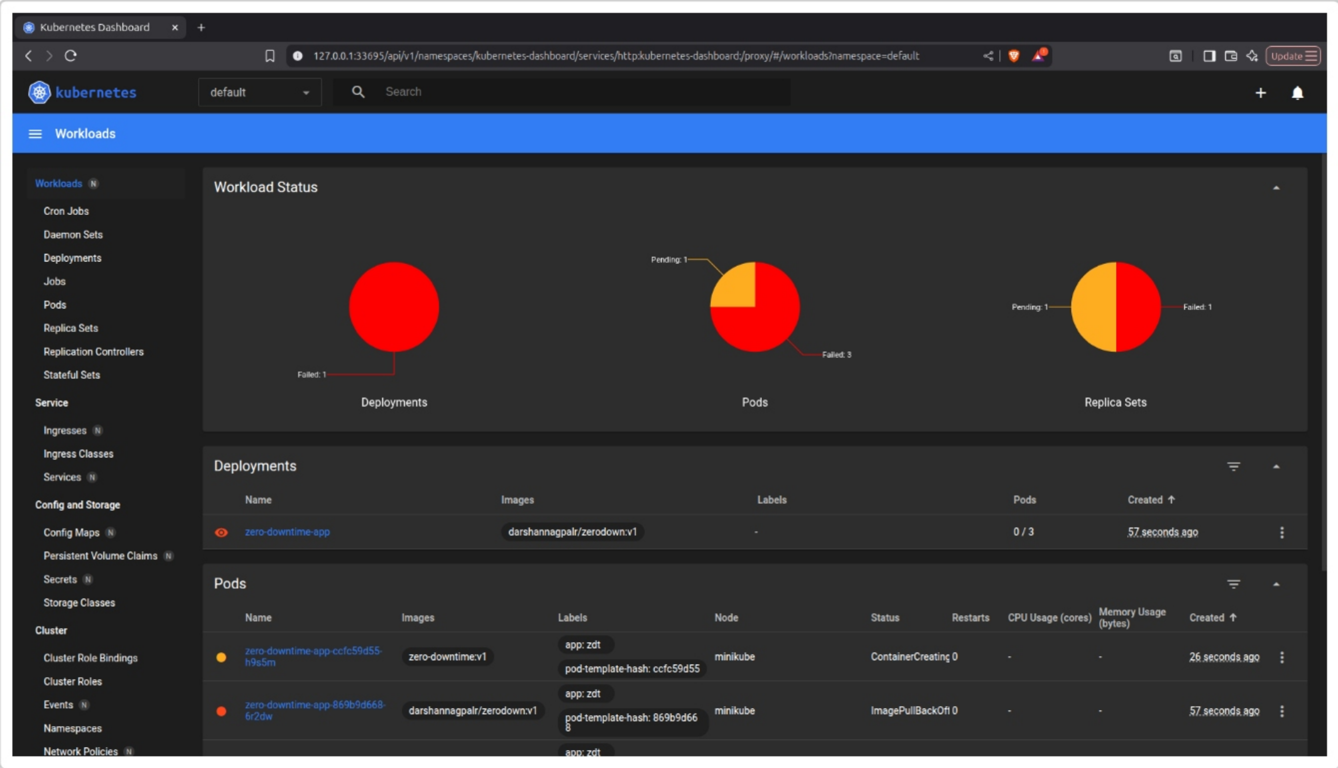
```
1rv24mc078_prateek_kumar@arch minikube service nginx-service
|-----------|---------------|-------------|---------------------------|
| NAMESPACE |     NAME      | TARGET PORT |            URL            |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:32649 |
|-----------|---------------|-------------|---------------------------|

Opening service default/nginx-service in default browser...
```

Step 8: Launch Kubernetes Dashboard

```
1rv24mc078_prateek_kumar@arch minikube dashboard
🔎  Verifying dashboard health...
🚀  Launching proxy...
🌐  Opening http://127.0.0.1:xxxxx/api/v1/namespaces/kubernetes-
dashboard/services/http:kubernetes-dashboard:/proxy/
```

Output

Nginx Application and Kubernetes Dashboard:



Result

The Nginx application was successfully deployed and exposed using a NodePort service. The Kubernetes Dashboard was launched using the CLI, allowing visualization and monitoring of cluster resources.